

Inputs

Query q

Context c

Trace $z_1 : \kappa$

Forward passes

$$p_{\theta}(y \mid q)$$

$$p_{\text{ctx}}(y \mid q, c)$$

$$p(y \mid \text{trace})$$

Reliability plug-in

Signals $s(q, c)$
(BM25, pop., entropy)

$$\hat{w}(c) = \hat{\mathbb{E}}[r \mid s]$$

Bayes-optimal combination

$$p^* \propto p_{\theta}^{1-\hat{w}} p_{\text{ctx}}^{\hat{w}} \text{ (T1)}$$

$$p_{\eta} \propto p^{\eta} \text{ (\eta-temp., T3)}$$

Decision

$$\hat{y} = \arg \max p$$